```
sharanteja2211@SharanTeja:~$ mkdir project
sharanteja2211@SharanTeja:~$ echo "Linux DevOps practice" > project/notes.txt
sharanteja2211@SharanTeja:~$ echo "Backup and compression practice" > project/readme.txt
sharanteja2211@SharanTeja:~$ ls
Demo  filDemo  file1.txt  file2.txt  marks.txt  notes.txt  part_aa  part_ab  project  students.txt
sharanteja2211@SharanTeja:~$ cd project/
sharanteja2211@SharanTeja:~/project$ ls
notes.txt  readme.txt
sharanteja2211@SharanTeja:~/project$ cat notes.txt
Linux DevOps practice
sharanteja2211@SharanTeja:~/project$ cat readme.txt
Backup and compression practice
sharanteja2211@SharanTeja:~/project$ cd..
cd..: command not found
sharanteja2211@SharanTeja:~/project$ cd ..
sharanteja2211@SharanTeja:~$ tar -cvf project.tar project/
project/
project/readme.txt
project/notes.txt
sharanteja2211@SharanTeja:~$ cd
sharanteja2211@SharanTeja:~$ ls
Demo  filDemo  file1.txt  file2.txt  marks.txt  notes.txt  part_aa  part_ab  project  project.tar  students.txt
sharanteja2211@SharanTeja:~$ ls -lh project.tar
-rw-r--r-- 1 sharanteja2211 sharanteja2211 10K Oct 27 09:23 project.tar
sharanteja2211@SharanTeja:~$ tar -tf project.tar
project/
project/readme.txt
project/notes.txt
sharanteja2211@SharanTeja:~$ tar -tf project.tar
project/
project/readme.txt
project/notes.txt
sharanteja2211@SharanTeja:~$ mkdir extracted
sharanteja2211@SharanTeja:~$ tar -xvf project.tar -C extracted/
project/
project/readme.txt
project/notes.txt
```

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sharanteja2211@SharanTeja:~$ tar -xvf project.tar -C extracted/
project/
project/readme.txt
project/notes.txt
sharanteja2211@SharanTeja:~$ tar -xzvf project.tar.gz
tar (child): project.tar.gz: Cannot open: No such file or directory
tar (child): Error is not recoverable: exiting now
tar: Child returned status 2
tar: Error is not recoverable: exiting now
sharanteja2211@SharanTeja:~$ gzip project.tar
sharanteja2211@SharanTeja:~$ ls -lh
total 48K
drwxr-xr-x 2 sharanteja2211 sharanteja2211 4.0K Oct 12 10:13 Demo
drwxr-xr-x 3 sharanteja2211 sharanteja2211 4.0K Oct 27 09:24 extracted
-rw-r--r-- 1 sharanteja2211 sharanteja2211   16 Oct  9 10:10 filDemo
-rw-r--r-- 1 sharanteja2211 sharanteja2211   13 Oct 22 09:10 file1.txt
-rw-r--r-- 1 sharanteja2211 sharanteja2211   14 Oct 22 09:10 file2.txt
-rw-r--r-- 1 sharanteja2211 sharanteja2211   14 Oct 22 09:48 marks.txt
-rw-r--r-- 1 sharanteja2211 sharanteja2211   20 Oct 22 09:07 notes.txt
-rw-r--r-- 1 sharanteja2211 sharanteja2211   13 Oct 22 09:10 part_aa
-rw-r--r-- 1 sharanteja2211 sharanteja2211    7 Oct 22 09:10 part_ab
drwxr-xr-x 2 sharanteja2211 sharanteja2211 4.0K Oct 27 09:22 project
-rw-r--r-- 1 sharanteja2211 sharanteja2211  240 Oct 27 09:23 project.tar.gz
-rw-r--r-- 1 sharanteja2211 sharanteja2211   18 Oct 22 09:48 students.txt
sharanteja2211@SharanTeja:~$ gzip -k project.tar
gzip: project.tar: No such file or directory
sharanteja2211@SharanTeja:~$ tar -cvf project.tar project/
project/
project/readme.txt
project/notes.txt
sharanteja2211@SharanTeja:~$ gzip -k project.tar
gzip: project.tar.gz already exists; do you wish to overwrite (y or n)? y
sharanteja2211@SharanTeja:~$ ls -lh
total 60K
drwxr-xr-x 2 sharanteja2211 sharanteja2211 4.0K Oct 12 10:13 Demo
drwxr-xr-x 3 sharanteja2211 sharanteja2211 4.0K Oct 27 09:24 extracted
-rw-r--r-- 1 sharanteja2211 sharanteja2211   16 Oct  9 10:10 filDemo
```

```
sharanteja2211@SharanTeja:~$ gzip -dk project.tar.gz
gzip: project.tar already exists; do you wish to overwrite (y or n)? y
sharanteja2211@SharanTeja:~$ echo "This is a gzip test" > sample.txt
gzip sample.txt
sharanteja2211@SharanTeja:~$ gunzip sample.txt.gz
sharanteja2211@SharanTeja:~$ gzip -c sample.txt > sample.txt.gz
sharanteja2211@SharanTeja:~$ zcat sample.txt.gz
This is a gzip test
sharanteja2211@SharanTeja:~$ zip -r project.zip project/
Command 'zip' not found, but can be installed with:
sudo apt install zip
sharanteja2211@SharanTeja:~$ sudo apt install zip
[sudo: authenticate] Password:
Installing:
  zip

Installing dependencies:
  unzip

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 0
  Download size: 355 kB
  Space needed: 988 kB / 1024 GB available

Continue? [Y/n]
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 unzip amd64 6.0-29ubuntu1 [180 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 zip amd64 3.0-15ubuntu3 [175 kB]
Fetched 355 kB in 1s (242 kB/s)
Selecting previously unselected package unzip.
(Reading database ... 35918 files and directories currently installed.)
Preparing to unpack .../unzip_6.0-29ubuntu1_amd64.deb ...
Unpacking unzip (6.0-29ubuntu1) ...
Selecting previously unselected package zip.
Preparing to unpack .../zip_3.0-15ubuntu3_amd64.deb ...
Unpacking zip (3.0-15ubuntu3) ...
Setting up unzip (6.0-29ubuntu1) ...
```

```
sharanteja2211@SharanTeja:~$ zip -r project.zip project/
  adding: project/ (stored 0%)
  adding: project/readme.txt (stored 0%)
  adding: project/notes.txt (stored 0%)
sharanteja2211@SharanTeja:~$ zip -r project.zip project/
updating: project/ (stored 0%)
updating: project/readme.txt (stored 0%)
updating: project/notes.txt (stored 0%)
sharanteja2211@SharanTeja:~$ unzip -l project.zip
Archive:  project.zip
  Length      Date    Time    Name
---------  ---------- -----   ----
        0  2026-10-27 09:22   project/
       32  2026-10-27 09:22   project/readme.txt
       22  2026-10-27 09:22   project/notes.txt
---------                     -------
       54                     3 files
sharanteja2211@SharanTeja:~$ unzip project.zip -d zip_output/
Archive:  project.zip
   creating: zip_output/project/
 extracting: zip_output/project/readme.txt
 extracting: zip_output/project/notes.txt
sharanteja2211@SharanTeja:~$ unzip project.zip -d zip_output/
Archive:  project.zip
replace zip_output/project/readme.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: y
 extracting: zip_output/project/readme.txt
replace zip_output/project/notes.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: y
 extracting: zip_output/project/notes.txt
sharanteja2211@SharanTeja:~$ echo "Linux compression practice" > sample2.txt
sharanteja2211@SharanTeja:~$ bzip2 sample2.txt
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sharanteja2211@SharanTeja:~$ bzip2 sample2.txt
sharanteja2211@SharanTeja:~$ xz -k sample3.txt
xz: sample3.txt: No such file or directory
sharanteja2211@SharanTeja:~$ xz -k sample3.txt
xz: sample3.txt: No such file or directory
sharanteja2211@SharanTeja:~$ mkdir -p backup
sharanteja2211@SharanTeja:~$ rsync -av project/ backup/
sending incremental file list
./
notes.txt
readme.txt

sent 269 bytes  received 57 bytes  652.00 bytes/sec
total size is 54  speedup is 0.17
sharanteja2211@SharanTeja:~$ rsync -av project backup/
sending incremental file list
project/
project/notes.txt
project/readme.txt

sent 287 bytes  received 58 bytes  690.00 bytes/sec
total size is 54  speedup is 0.16
sharanteja2211@SharanTeja:~$ rsync -avn project/ backup/
sending incremental file list
./

sent 129 bytes  received 19 bytes  296.00 bytes/sec
total size is 54  speedup is 0.36 (DRY RUN)
```